

$$x = -0.1101, y = -0.1011$$

等价 $x' = +0.1101, y' = +0.1011$ 相乘

(补码) 被乘数

乘数

部分积

00.1101

00.1011:0

00.0000

00.0000

① 10: $+[-x']$, 右移

$+[-x']$ 11.0011

11.0011

0.1011:0

→ 11.1001 1.0101:1

② 11: 右移

→ 11.1100 1.1010:1

③ 01: $+ [x']$, 右移

$+ [x']$ 00.1101

00.1001

1.1010:1

→ 00.0100 1.1101:0

④ 10: $+ [-x']$, →

$+ [-x']$ 11.0011

11.0111

1.1101:0

→ 11.1011 1.1110:1

⑤ 01: $+ [x']$

$+ [x']$ 00.1101

00.0000

1.1110:1

$$P = x \cdot y = x' \cdot y' = 0.10001111$$

$n=4$, 重复 n 次

$x = -0.1011$, $y = -0.1101$, x/y 等价于 $x' = +0.1011$, $y = +0.1101$, x'/y'

被除数/余数

$$[-y]_{补} = 11.0011$$

00.1011

$$\begin{array}{r} +[-y]_{补} \ 11.0011 \\ \hline 11.1110 \end{array} \quad \begin{array}{l} \text{商} \\ 0 \end{array}$$

左移 11.1100

$$\begin{array}{r} +[y]_{补} \ 00.1101 \\ \hline 00.1001 \end{array} \quad 1$$

左移 01.0010

$$\begin{array}{r} +[-y]_{补} \ 11.0011 \\ \hline 00.0101 \end{array} \quad 1$$

左移 00.1010

$$\begin{array}{r} +[y]_{补} \ 00.1101 \\ \hline 11.1101 \end{array} \quad 0$$

左移 1.110

左移 11.1010

$$\begin{array}{r} +[y]_{补} \ 00.1101 \\ \hline 00.0111 \end{array} \quad 1$$

$$[x/y]_{补} = 0.1101, \text{ 余 } 0.0111 \times 2^{-4}$$